



DSP1 – Practice Homework

Exercise 1. Review of complex numbers.

- (a) Let $s[n] = \frac{1}{2^n} + j\frac{1}{3^n}$. Compute $\sum_{n=1}^{\infty} s[n]$.
- (b) Do the same with $s[n] = \left(\frac{j}{3}\right)^n$.
- (c) Characterize the set of complex numbers satisfying $z^* = z^{-1}$.
- (d) Find 3 distinct complex numbers $\{z_0, z_1, z_2\}$ for which $z_i^3 = 1$.
- (e) Compute the infinite product $\prod_{n=1}^{\infty} e^{j\pi/2^n}$

Exercise 2. Sampling music

A music song recorded in a studio is stored as a digital sequence on a CD. The analog signal representing the music is 2 minutes long and is sampled at a frequency $f_s = 44100 \text{ s}^{-1}$. How many samples should be stored on the CD?

Exercise 3. Moving average

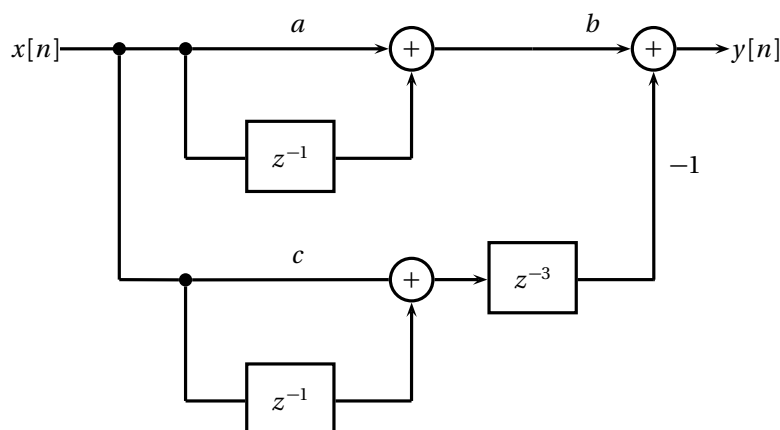
Consider the following signal,

$$x[n] = \delta[n] + 2\delta[n-1] + 3\delta[n-2]. \quad (1)$$

Compute its moving average $y[n] = \frac{x[n] + x[n-1]}{2}$, where we call $x[n]$ the input and $y[n]$ the output.

Exercise 4. SP with Lego

Given the following filter:



What is the input-output relationship?

Exercise 5. Bases

Let $\{\mathbf{x}^{(k)}\}_{k=0,\dots,N-1}$ be a basis for a subspace S . Prove that any vector $\mathbf{z} \in S$ is *uniquely* represented in this basis. *Hint: remember that the vectors in a basis are linearly independent and use this to prove the thesis by contradiction.*

Exercise 6. Vector Spaces.

Consider the four diagonals of a three-dimensional unit cube as vectors in $\mathbb{R}^3$. Are they mutually orthogonal?

Exercise 7. DFT Formula.

Derive a simple expression for the DFT of the time-reversed signal

$$\mathbf{x}_r = [x[N-1] \ x[N-2] \ \dots \ x[1] \ x[0]]^T$$

in terms of the DFT $\mathbf{X}$ of the signal $\mathbf{x}$. Hint: you may find it useful to remark that $W_N^k = W_N^{-(N-k)}$.

Exercise 8. DFT Manipulation.

Consider a length- N signal $\mathbf{x} = [x[0] \ x[1] \ \dots \ x[N-1]]^T$ and the corresponding vector of DFT coefficients $\mathbf{X} = [X[0] \ X[1] \ \dots \ X[N-1]]^T$.

Consider now the length- $2N$ signal obtained by interleaving the values of $\mathbf{x}$ with zeros

$$\mathbf{x}_2 = [x[0] \ 0 \ x[1] \ 0 \ x[2] \ 0 \ \dots \ x[N-1] \ 0]^T$$

Express $\mathbf{X}_2$ (the $2N$ -point DFT of $\mathbf{x}_2$) in terms of $\mathbf{X}$.

Exercise 9. The Structure of DFT Formulas.

The DFT and IDFT formulas are similar, but not identical. Consider a length- N signal $\mathbf{x}$ with $x[n], N = 0, \dots, N-1$. What is the length- N signal $y[n]$ obtained as

$$\mathbf{y} = \text{DFT}\{\text{DFT}\{\mathbf{x}\}\}?$$

In other words, what are the effects of applying twice the DFT transform?

Exercise 10. DFT of the Autocorrelation.

Consider a sequence $\mathbf{x}$ of finite length N . Let $\mathbf{X}$ denote the N point DFT of $\mathbf{x}$ and define the circular autocorrelation sequence $\mathbf{r}_x$ as

$$r_x[m] = \sum_{n=0}^{N-1} x[n] x^*[(n-m) \bmod N].$$

Express $\mathbf{r}_x$ in terms of $\mathbf{X}$. [Hint: build a signal $S[n] = X[n] X^*[n]$, compute its inverse DFT and work backwards.]

Exercise 11. Subsampling the DFT

Consider $\mathbf{x} \in \mathbb{C}^N$ with N even and its N -point DFT $\mathbf{X}$. Define an $(N/2)$ -length vector $\mathbf{Y}$ as $Y[k] = X[2k], k = 0, \dots, N/2-1$. Compute the inverse DFT of $\mathbf{Y}$.

Exercise 12. Plancherel-Parseval Equality.

Let $x[n]$ and $y[n]$ be two complex-valued sequences and $X(e^{j\omega})$ and $Y(e^{j\omega})$ their corresponding DTFTs.

(a) Show that

$$\langle x[n], y[n] \rangle = \frac{1}{2\pi} \langle X(e^{j\omega}), Y(e^{j\omega}) \rangle,$$

where we use the inner products for $l_2(\mathbb{Z})$ and $L_2([-\pi, \pi])$ respectively.

(b) What is the physical meaning of the above formula when $x[n] = y[n]$?

Exercise 13. DTFT properties.

Derive the time-reverse and time-shift properties of the DTFT.
